

Initial
library

1

$$\lambda_0 = \{\lambda_0(s)\}_{s \in S}$$

(deterministic)

sampling

2

$$\lambda_1 \sim \text{Poisson}(\mu_1)$$

production
of capsid

3

$$\lambda_2 = \mu_2$$

selectivity_step
(selectivity)

4

$$\lambda_3(s) = \lambda_2(s) \cdot \exp \frac{\text{score}_{\text{sel}}(s)}{T_{\text{sel}}}$$

NGS_approx
(first NGS)

1'

$$\lambda'_0 \sim \text{NB}(r_1, p_{1,s})$$
$$\left(D, \left\{ \frac{\lambda_0^{\text{PCR}}(s)}{\sum_u \lambda_0^{\text{PCR}}(u)} \right\}_{s \in S} \right)$$

NGS_approx
(second NGS)

3'

$$\lambda'_2 \sim \text{NB}(r_2, p_{4,s})$$
$$\left(D, \left\{ \frac{\lambda_2^{\text{PCR}}(s)}{\sum_u \lambda_2^{\text{PCR}}(u)} \right\}_{s \in S} \right)$$

NGS_approx
(third NGS)

4'

$$\lambda'_3 \sim \text{NB}(r_3, p_{6,s})$$
$$\left(D, \left\{ \frac{\lambda_3^{\text{PCR}}(s)}{\sum_u \lambda_3^{\text{PCR}}(u)} \right\}_{s \in S} \right)$$

Initial library

Variables :

S : set of all sequences

D : sequencing depth

N_1 : molecules sampled

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$s \in S$
 $c)$

sample_approx (sampling)

Parameters (Poisson) :

$$\mu_1 = N_1 \cdot \frac{\lambda_0(s)}{\sum_u \lambda_0(s)}$$

sampling

2

$$\lambda_1 \sim \text{Poisson}(\mu_1)$$

production
of capsid

$$\lambda_2 \sim \text{Poisson}(\mu_2)$$

production of capsids

For each sequence $s \in S$:

Number of transfected cells :

$$C_s \sim \text{Poisson}(\rho \cdot \lambda_2(s))$$

For each cell $j = 1, \dots, C_s$:

$$E_{s,j} = \exp(\varepsilon Z_{s,j}), \quad Z_{s,j} \sim \mathcal{N}(0, 1)$$

$$V_{s,j} \sim \text{Poisson}\left(\alpha E_{s,j} \exp\left(\frac{\text{score}_{\text{viab}}(s)}{T_1}\right)\right)$$

(capsids produced by cell j)

$$\lambda_2(s) \approx \mu_2(s) = \sum_{j=1}^{C_s} V_{s,j}$$

$\mu_1)$

$$\text{score}_{\text{viab}}(s) = \sum_{i=1}^L F_i^{(1)}(s_i) + \sum_{1 \leq j < i \leq L} J_{i,j}^{(1)}(s_i, s_j)$$

production of
capsids

3

$$\lambda_2 = \mu_2$$

selectivity_ste
(selectivity)

$\lambda_3(s)$

NGS_approx
(second NGS)

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$$\lambda'_2 \sim \text{NB}(r_2, p_{4,s})$$

$$\left(D, \left\{ \frac{\lambda_2^{\text{PCR}}(s)}{\sum_u \lambda_2^{\text{PCR}}(u)} \right\}_{s \in S} \right)$$

selectivity_step
(selectivity)

4

$$\lambda_3(s) = \lambda_2(s) \cdot \exp \frac{\text{score}_{sel}(s)}{T_{sel}}$$

NGS_approx
(third NGS)

4'

$$\lambda'_3 \sim \text{NB}(r_3, p_{6,s})$$
$$\left(D, \left\{ \frac{\lambda_3^{\text{PCR}}(s)}{\sum_u \lambda_3^{\text{PCR}}(u)} \right\}_{s \in S} \right)$$

Selectivity

$$\text{score}_{sel}(s) = \sum_{i=1}^L F_i^{(2)}(s_i) + \sum_{1 \leq j < i \leq L} J_{i,j}^{(2)}(s_i, s_j)$$